Inspiring Innovative Ideas

COLLEGE OF ENGINEERING TRIKARIPUR

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DEPARTMENT OF COMPUTER SCIENCE

FOUNDATIONS OF COMPUTING: FROM HARDWARE ESSENTIALS TO WEB DESIGN

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MODULE 4

- Web Design- Understanding the web content delivery
- Understanding HTML and XHTML Connections
- Understanding Cascading Style Sheets
- Understanding JavaScript

Publishing web Page Content

- Web servers make your content available to others who, in turn, use their web browsers to navigate to an address and wait for the server to send information to them.

- You will be intimately involved in this publishing process because you must create files and then put them on a server to make them available in the first place, and you must ensure that your content will appear to the end user as you intended.

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A Brief History of HTML and the World Wide Web

- Until 1990, accessing information through the Internet was a rather technical affair. It was so hard, in fact, that even Ph.D.-holding physicists were often frustrated when trying to swap data.

- One such physicist, the now famous (and knighted) Sir Tim Berners-Lee, cooked up a way to easily cross-reference text on the Internet through hypertext links.

- Hypertext originally meant text stored in electronic form with cross-reference links between pages.
- It is now a broader term that refers to just about any object (text, images, files, and so on) that can be linked to other objects.
- **Hypertext Markup Language** is a language for describing how text, graphics, and files containing other information are organized and linked together.

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A Brief History of HTML and the World Wide Web

- By 1993, only 100 or so computers throughout the world were equipped to serve up HTML pages.
- Those interlinked pages were dubbed the World Wide Web (WWW), and several web browser programs had been written to allow people to view web pages.
- Because of the growing popularity of the Web, a few programmers soon wrote web browsers that could view graphical images along with text.
- The continued development of web browser software and the standardization of the HTML— and XHTML—languages has lead us to the world we live in today, one in which more than 110 million web servers answer requests for more than 25 billion text and multimedia files.

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Creating Web Content

- Web content is **the text, visual or audio content that is made available online and user encountered as part of the online usage and experience on websites**. It may include text, images, sounds and audio, online videos, among other items placed within web pages.

- “visiting a web page,” what we really mean is something like “looking at all the text and the images at one address on our computer.”

- The text that we read, and the images that we see, are rendered by our web browsers, which are given certain instructions found in individual files.

- Those files contain text that is marked up, or surrounded by, HTML code that tell the browser how to display the text—as a heading, as a paragraph, in a red font, and so on.

Understanding Web Content Delivery

- There are several steps in the process—and potentially several trips between the browser and server—before you see the entire content of the site you requested.

Understanding Web Content Delivery

- Suppose you want to do a Google search, so you dutifully type http://www.google.com in the address bar or select the Google bookmark from your bookmarks list.

Understanding Web Content Delivery

- Figure shows a website that contains text plus one image (the Google logo).
- A simple version of the processes that occurred to retrieve that text and image from a web server and display it on your screen is as follows:

1. Your web browser sends a request for the index.html file located at the **http://www.google.com/** address. The **index.html** file does not have to be part of the address that you type in the address bar.
2. After receiving the request for a specific file, the web server process looks in its directory contents for the specific file, opens it, and sends the content of that file back to your web browser.
3. The web browser receives the content of the index.html file, which is text marked up with HTML codes, and renders the content based on these HTML codes.

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Understanding Web Content Delivery

Eg : <img src =”/logos/logo.gif” width=”384” height=”121” border =”0”

alt= ”Google”/>

4. The browser looks at the src attribute in the <img/> tag to find th source location. In this case, the image logo.gif can be found in the **logos directory** at the same web address (www.google.com) from which the browser retrieved the HTML file.

5. The browser requests the file at the http://www.google.com/logos/logo.gif web address.

6. The web server interprets that request, finds the file, and sends the contents of that file to the web browser that requested it.

7. The web browser displays the image on your monitor.

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Selecting a Web Hosting Provider

- If you type **web hosting provider** in your search engine of choice, you will get millions of hits and an endless list of sponsored search results.
- Some selection criteria for a web hosting provider include the following,

1. Reliability/server "uptime"—If you have an online presence, you want to make sure people can actually get there consistently.
2. Customer service—Look for multiple methods for contacting customer service (phone, email, and chat) as well as online documentation for common issues.
3. Server space—Does the hosting package include enough server space to hold all the multimedia files (images, audio, and video).
4. Bandwidth—Does the hosting package include enough bandwidth so that all the people visiting your site and downloading files can do

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Selecting a Web Hosting Provider

5. Domain name purchase and management—Does the package include a custom domain name, or must you purchase and maintain your domain name separately from your hosting account?

6. Price—Do not overpay for hosting. You will see a wide range of price offered and should immediately wonder “what’s the difference?

Often the difference has little to do with the quality of the service an everything to do with company overhead and what the company thinks they can get away with charging people.

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Testing with Multiple Web Browsers

You should always test your websites with as many of these web browsers as possible:

- Apple Safari (http://www.apple.com/safari/) for Mac and Windows
- Google Chrome (http://www.google.com/chrome) for Windows
- Mozilla Firefox (http://www.mozilla.com/firefox/) for Mac, Windows, and Linux
- Microsoft Internet Explorer (http://www.microsoft.com/ie) for Windows
- Opera (http://www.opera.com/) for Mac, Windows, and Linux/UNIX

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Using FTP to Transfer Files

- You have to put your web content on a web server to make it accessible to others. This process typically occurs by using File Transfer Protocol (FTP). To use FTP, you need an FTP client—a program used to transfer files from your computer to a web server.
- FTP clients require three pieces of information to connect to your web server; this information will have been sent to you by your hosting provider after you set up your account:

1. The hostname, or address, to which you will connect
2. Your account username
3. Your account password

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Selecting an FTP Client

- Figure shows an example of FireFTP, which is an FTP client used with the Firefox web browser.

Using FTP to Transfer Files

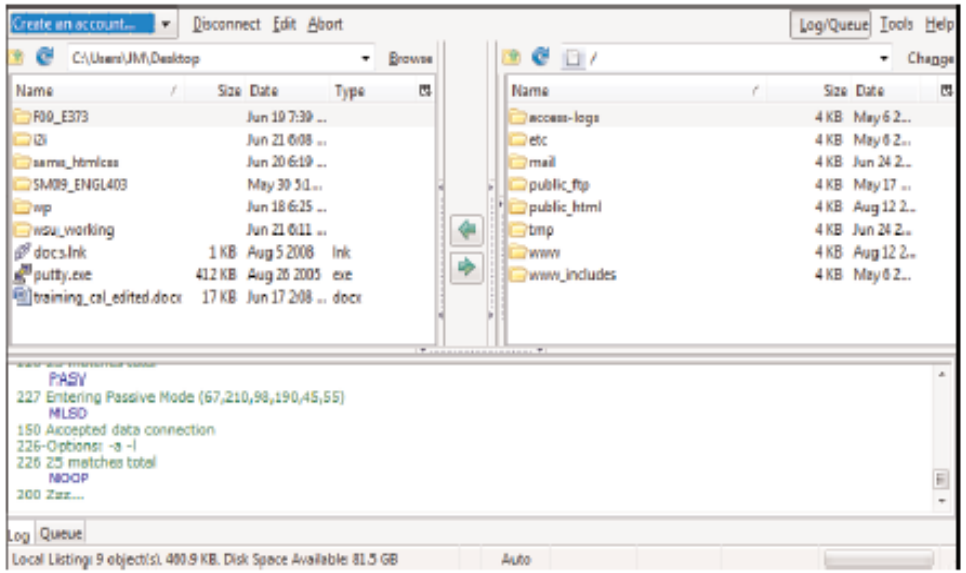

FIGURE 1.4
The FireFTP interface.

Selecting an FTP Client

- The directory listing of the local machine (your computer) appears on the left of your screen and the directory listing of the remote machine (the web server) appears on the right.
- Typically, you will see right-arrow and left-arrow buttons—as shown in Figure
- The right arrow sends selected files from your computer to your web server; the left arrow sends files from the web server to your computer.
- Many FTP clients also enable you to simply select files, and then drag and drop those files to the target machines.

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Selecting an FTP Client

Here are some popular free FTP clients:

- Classic FTP (http://www.nchsoftware.com/classic/) for Mac and Windows
- Cyberduck (http://cyberduck.ch/) for Mac
- Fetch (http://fetchsoftworks.com/) for Mac
- FileZilla (http://filezilla-project.org/) for all platforms
- FireFTP (http://fireftp.mozdev.org/) Firefox extension for all platforms

- When you have selected an FTP client and installed it on your computer, you are ready to upload and download files from your web server.

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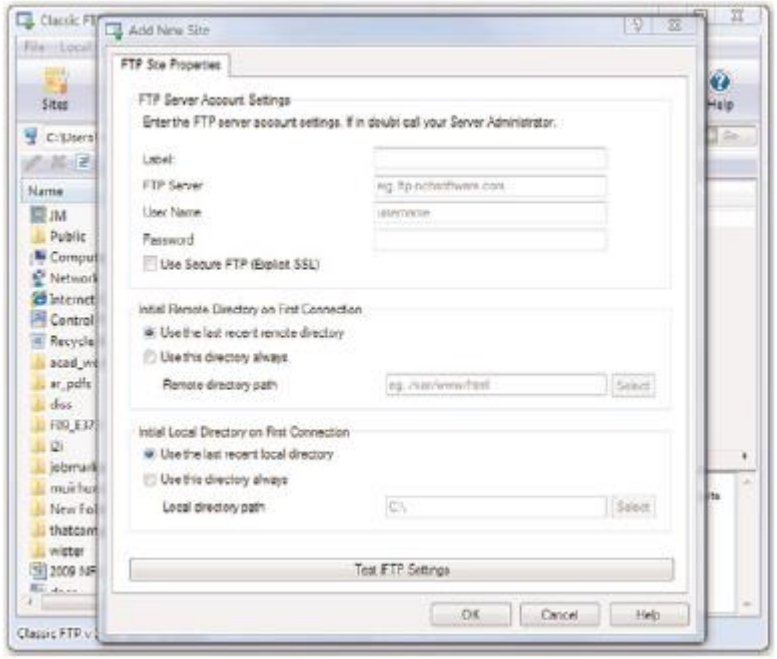
Using an FTP Client

The following steps show how to use Classic FTP to connect to your we server and transfer a file. you first need the hostname, the account username, and the account password.

1. Start the Classic FTP program and click the Connect button. You will be prompted to fill out information for the site to which you want to connect, as shown in Figure.



Using an FTP Client

2. Fill in each of the items shown in Figure as follows:

- The site Label is the name you'll use to refer to your own site. Nobody else will see this name, so enter whatever you want.
- The FTP Server is the FTP address of the web server to which you need to send your web pages. This address will have been given to you by your hosting provider. It will probably be yourdomain.com, but check the information you received when you signed up for service.
- The User Name field and the Password field should also be completed using information given to you by your hosting provider.
- Don't change the values for Initial Remote Directory on First Connection and Initial Local Directory on First Connection until you are used to using the client and have established a workflow.

Using an FTP Client

3. When you're finished with the settings, click OK to save the settings and establish a connection with the web server.

- You will see a dialog box indicating that Classic FTP is attempting to connect to the web server. Upon successful connection, you will see an interface similar to Figure, showing the contents of the local directory on the left and the contents of your web server on the right.

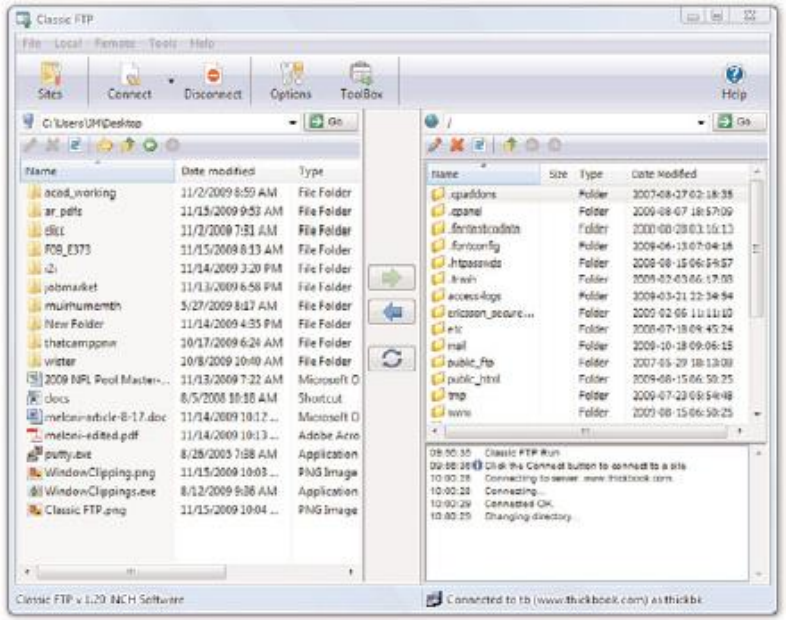


FIGURE 1.6
A successful connection to a remote web server via Classic FTP.

Using an FTP Client

4. You are now almost ready to transfer files to your web server. All the remains is to change directories to what is called the document root of your web server. The document root of your web server is the directory that is designated as the top-level directory for your web content the starting point of the directory structure . This directory will be name public_html , www , or htdocs. This is not a directory that you will have to create because your hosting provider will have created it for you.

5. The goal is to transfer the sample.html file you created earlier from your computer to the web server.

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Using an FTP Client

6. Click the right-arrow button in the middle of the client interface to send the file to the web server. After the file transfer is completed the right side of the client interface should refresh to show you that the file has made it to its destination.

7. Click the Disconnect button to close the connection, and then exit out of the Classic FTP program

Distributing Content Without a Web Server

Publishing Content Locally

- You need to create a local site that you want to distribute on USB drive. Even the cheapest USB drives hold so much data these days and basic hypertext files are quite small—that you can distribute an entire site and a fully functioning web browser all on one little drive.
- The top-level of the USB drive directory structure can be your document root. Or if you are distributing a web browser along with the content, you might have two directories—for example, one named browser and one named content.
- Within the document root, you could have additional subfolders in which you place content and other multimedia assets.

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Publishing Content on a Blog

- You might have a blog hosted by a third-party, such as Blogger or WordPress (among others), and thus have already published content without having a dedicated web server or even knowing any HTML.

- These services offer visual editors in addition to source editors, meaning that you can type your words and add visual formatting such as bold, italics, or font colors without knowing the HTML for these actions.

- The content becomes actual HTML when you click the Publish button in these editors.

Understanding HTML and XHTML Connections

Getting Prepared

- Get a computer, with Ubuntu (Linux), Windows, Macintosh

Getting Started with a Simple Web Page

- To create these text files, use a *text editor* such as Notepad (on Windows) or TextEdit (on a Mac)—do not use WordPad, Microsoft Word, or other full-featured word-processing software because those create different sorts of files than the plain-text files we use for web content.

You should start with some text that you want to put on a web page:

1. Find (or write) a few paragraphs of text about yourself, your family, your company, your softball team, or some other subject in which you're interested.
2. Save this text as plain, standard ASCII text.

- When you save files containing HTML tags, always give them a name ending in .html or .htm, which is also an acceptable file extension to use.

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HTML—Hyper Text Markup Language

- Every web page you create must include the

  <html> </html>, <head> </head>, <title> </title>, and <body> </body> tagpairs.

<!-- Write your comments here --> comment in html

HTML—Hyper Text Markup Language

- Most HTML tags have two parts: an *opening tag*, which indicates where a piece of text begins, and a *closing tag*, which indicates where the piece of text ends. Closing tags start with a / (forward slash) just after the < symbol.
- Another type of tag is the *empty tag*, which is unique in that it doesn't include a pair of matching opening and closing tags. Instead, an empty tag consists of a single tag that starts with a < and ends with a / just before the > symbol.

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HTML—Hyper Text Markup Language

- An *opening tag* is an HTML tag that indicates the start of an HTML command; the text affected by the command appears after the opening tag. Opening tags always begin with < and end with >, as in <html>.

- A *closing tag* is an HTML tag that indicates the end of an HTML command; the text affected by the command appears before the closing tag. Closing tags always begin with </ and end with >, as in </html>.

- An *empty tag* is an HTML tag that issues an HTML command without enclosing any text in the page. Empty tags always begin with < and end with />, as in <br /> and <img />.

HTML—Hyper Text Markup Language

- For example, the <body> tag tells the web browser where the actual body text of the page begins, and </body> indicates where it ends.
- Everything between the <body> and </body> tags will appear in the main display area of the web browser window.
- title text, which is any text that is located between <title> and </title>. The <title> tag defines the title of the document. The title must be text-only, and it is shown in the browser's title bar or in the page's tab.
- Each section is identified by <head> and <body> tags.

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Structure of HTML

```
<html>
  <head>
    <title>Page title</title>
  </head>
  <body>
    <h1>This is a heading</h1>
    <p>This is a paragraph.</p>
    <p>This is another paragraph.</p>
  </body>
</html>
```

HTML Headings

HTML headings are defined with the <h1> to <h6> tags.

- <h1> defines the most important heading. <h6> defines the least important heading.
- <h1>Heading 1</h1>
- <h2>Heading 2</h2>
- <h3>Heading 3</h3>
- <h4>Heading 4</h4>
- <h5>Heading 5</h5>
- <h6>Heading 6</h6>

HTML Headings

```
<!DOCTYPE html>
<html>
<body>

<h1>Heading 1</h1>
<h2>Heading 2</h2>
<h3>Heading 3</h3>
<h4>Heading 4</h4>
<h5>Heading 5</h5>
<h6>Heading 6</h6>

</body>
</html>
```

HTML Headings - Output

Heading 1

Heading 2

Heading 3

Heading 4

Heading 5

Heading 6

HTML Paragraphs

- The HTML <p> element defines a paragraph.

- A paragraph always starts on a new line, and browsers automatically add some white space (a margin) before and after a paragraph.

HTML Paragraphs

```
<!DOCTYPE html>
<html>
<body>

<p>
This paragraph
contains a lot of lines
in the source code,
but the browser
ignores it.
</p>
```

HTML Paragraphs

```
<p>
This paragraph
contains      a lot of spaces
in the source     code,
but the    browser
ignores it.
</p>

<p>
The number of lines in a paragraph depends on the size of the browser window. If you resize the browser window, the
number of lines in this paragraph will change.
</p>
</body>
</html>
```

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HTML Paragraphs - Output

This paragraph contains a lot of lines in the source code, but the browser ignores it.

This paragraph contains a lot of spaces in the source code, but the browser ignores it.

The number of lines in a paragraph depends on the size of the browser window. If you resize the browser window, the number of lines in this paragraph will change.

<hr>,
 tags

- The <hr> element is most often displayed as a horizontal rule that is used to separate content (or define a change) in an HTML page.

- It is a self-closing tag.
- It's typically used to divide sections of content, such as headings, articles, or different topics.

- In HTML, the <br> tag inserts a line break, while the <hr> tag creates a horizontal rule or line to visually separate sections of content.
- It's a self-closing tag, meaning it doesn't require a closing tag (e.g., <br> or <br />).

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Horizontal ruler (hr tag) example

```
<!DOCTYPE html>
<html>
<body>

<h1>This is heading 1</h1>
<p>This is some text.</p>
<hr>
```

Horizontal ruler (hr tag) example….

```
<h2>This is heading 2</h2>
<p>This is some other text.</p>
<hr>

<h2>This is heading 2</h2>
<p>This is some other text.</p>

</body>
</html>
```

Output

This is heading 1

This is some text.

This is heading 2

This is some other text.

This is heading 2

This is some other text.

Line break(br tag) example

```
<!DOCTYPE html>
<html>
<body>

<p>This is<br>a paragraph<br>with line breaks.</p>

</body>
</html>
```

Output

This is
a paragraph
with line breaks.

Summary of html tags

Tag	Function
<html>...</html>	Encloses the entire HTML document.
<head>...</head>	Encloses the head of the HTML document. Used within the <html> tag pair.
<title>...</title>	Indicates the title of the document. Used within the <head> tag pair.
<body>...</body>	Encloses the body of the HTML document. Used within the <html> tag pair.
...	A paragraph; skips a line between paragraphs.
 	A line break.
<hr />	A horizontal rule line.
<h1>...</h1>	A first-level heading.
<h2>...</h2>	A second-level heading.
<h3>...</h3>	A third-level heading.
<h4>...</h4>	A fourth-level heading (seldom used).
<h5>...</h5>	A fifth-level heading (seldom used).
<h6>...</h6>	A sixth-level heading (seldom used).

HTML Lists

- HTML lists allow web developers to group a set of related items in lists.

Unordered HTML List

An unordered list starts with the <ul> tag. Each list item starts with the <li> tag.

The list items will be marked with bullets (small black circles) by default:

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Example

```
<!DOCTYPE html>
<html>
<body>

<h2>An unordered HTML list</h2>

<ul>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>

</body>
</html>
```

Output

An unordered HTML list

- Coffee
- Tea
- Milk

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HTML Lists

- **Ordered HTML List**

An ordered list starts with the <ol> tag. Each list item starts with the <li> tag.

The list items will be marked with numbers by default:

```
<!DOCTYPE html>
<html>
<body>

<h2>An ordered HTML list</h2>

<ol>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ol>

</body>
</html>
```

Output

An ordered HTML list

1. Coffee
2. Tea
3. Milk

HTML Text Formatting

Formatting elements were designed to display special types of text:

- <b> - Bold text
- - Important text
- <i> - Italic text
- - Emphasized text
- <mark> - Marked text
- <small> - Smaller text
- <del> - Deleted text
- <ins> - Inserted text
- <sub> - Subscript text
- <sup> - Superscript text

Example

```
<!DOCTYPE html>
<html>
<head>
   <title>Text Formatting Example</title>
</head>
<body>

   <p><b>Bold text</b></p>
   <p><strong>Important text</strong></p>
   <p><i>Italic text</i></p>
   <p><em>Emphasized text</em></p>
   <p><mark>Marked text</mark></p>
```

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Example

```
<p><small>Smaller text</small></p>
   <p><del>Deleted text</del></p>
   <p><ins>Inserted text</ins></p>
   <p>H<sub>2</sub>O (Subscript text example)</p>
   <p>10<sup>2</sup> (Superscript text example)</p>

</body>
</html>
```

Output

Bold text

Important text

Italic text

Emphasized text

Marked text

Smaller text

~~Deleted text~~

Inserted text

H_2O (Subscript text example)

10^2 (Superscript text example)

HTML Links

- HTML links are hyperlinks.
- You can click on a link and jump to another document.
- When you move the mouse over a link, the mouse arrow will turn into a little hand.

Note: A link does not have to be text. A link can be an image or any other HTML element!

- The HTML <a> tag defines a hyperlink. It has the following syntax:

```
<a href="url">link text</a>
```

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- The most important attribute of the <a> element is the href attribute, which indicates the link's destination.

- The link text is the part that will be visible to the reader.

- Clicking on the link text, will send the reader to the specified URL address.

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Example

```
<!DOCTYPE html>
<html>
<body>

<h1>HTML Links</h1>

<p><a href="https://www.ktu.edu.in/">Visit ktu.edu.in!</a></p>

</body>
</html>
```

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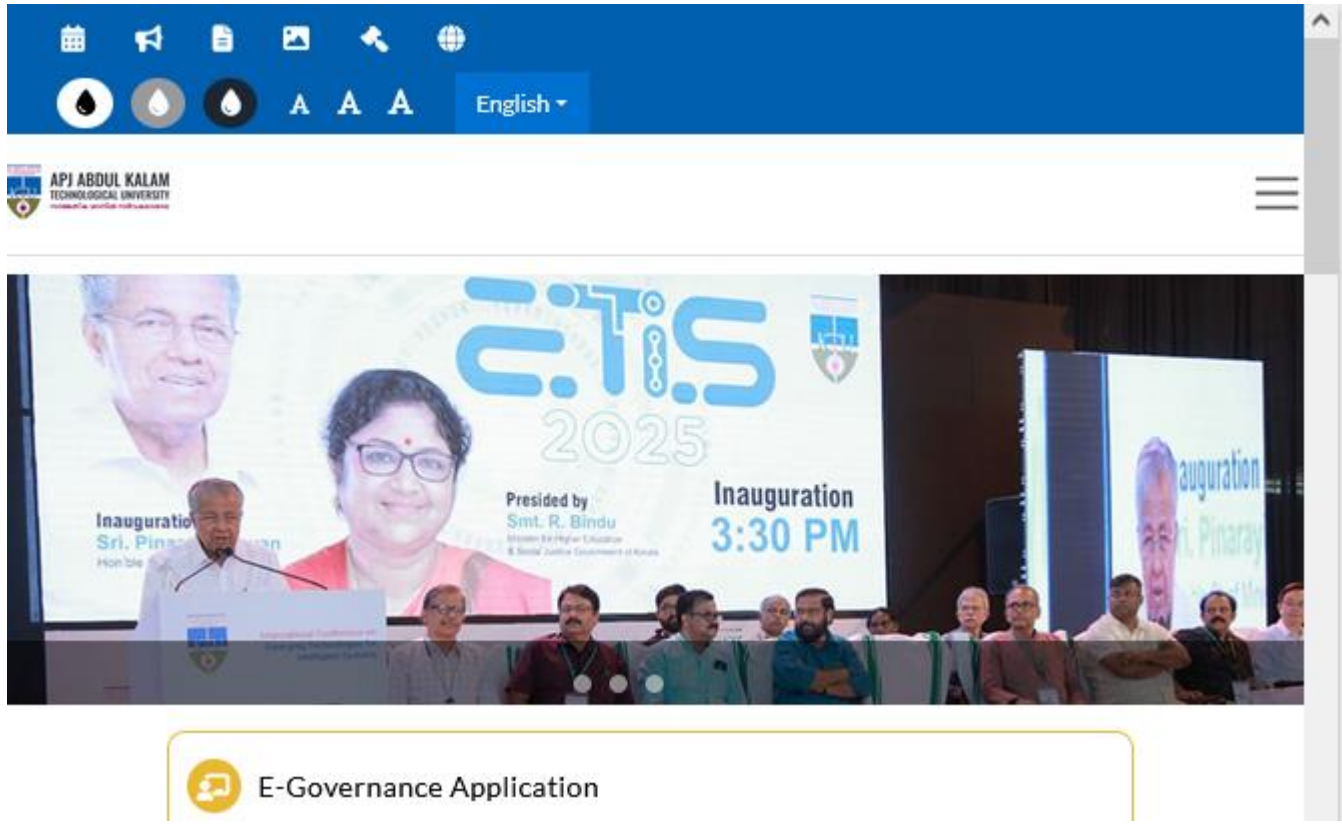
Output

HTML Links

Visit ktu.in!



HTML Images

- Images can improve the design and the appearance of a web page.

```
<img src="url" alt="alternatetext">
```

```
<!DOCTYPE html>
<html>
<body>

<h2>HTML Image</h2>
<img src=“img1.jpg" alt=“ROSE" width="500" height="333">

</body>
</html>
```

Output

HTML Image

div tag in HTML

- The tag defines a division or a section in an HTML document.
- The tag is used as a container for HTML elements - which is then styled with CSS or manipulated with JavaScript.
- The tag is easily styled by using the class or id attribute.
- Any sort of content can be put inside the tag!
- Note: By default, browsers always place a line break before and after the element.

Example

```
<!DOCTYPE html>
<html>
<html>
<head>
</head>
<body>
A div element is displayed like this:

<div><p>This is some text in a div element.</p></div>

</body>
</html>
```

Practice questions

What HTML tags and text would you use to produce the following web content:

- A small heading with the words We are Proud to Present
- A horizontal rule across the page
- A large heading with the one word Orbit
- A medium-sized heading with the words The Geometric Juggler
- Another horizontal rule

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Answer

```
<!DOCTYPE html>
<html>
<body>
<h3>We are Proud to Present</h3>
<hr >
<h1>Orbit</h1>
<h2>The Geometric Juggler</h2>
<hr >
</body>
</html>
```

Question 2

- What code would you use to create a complete HTML web page with the title Foo Bar, a heading at the top that reads Happy Hour at the Foo Bar, followed by the words Come on down! in regular type?

Answer

```
<!DOCTYPE  html>
<html>
<head>
<title>Foo Bar</title>
</head>
<body>
<h1>Happy Hour at the Foo Bar</h1>
<p>Come on Down!</p>
</body>
</html>
```

A Simple HTML Document

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>
</head>
<body>

<h1>My First Heading</h1>
<p>My first paragraph.</p>

</body>
</html>
```

A Simple HTML Document cont…

- The <!DOCTYPE html> declaration defines that this document is an HTML5 document.
- The <html> element is the root element of an HTML page
- The <head> element contains meta information about the HTML page
- The <title> element specifies a title for the HTML page (which is shown in the browser's title bar or in the page's tab)
- The <body> element defines the document's body, and is a container for all the visible contents, such as headings, paragraphs, images, hyperlinks, tables, lists, etc.
- The <h1> element defines a large heading
- The <p> element defines a paragraph

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XML

What is XML?

XML stands for **eXtensible Markup Language**.
It's a **text-based** format used to store and transport data in a **structured** and **readable** way.
It looks similar to HTML but is used for **data**, not **display**.

- Example of XML:

```
<student>
  <name>John Doe</name>
  <age>21</age>
  <major>Computer Science</major>
</student>
```

Here, data is stored inside custom tags like <name>, <age>, etc.

XHTML

Extensible Hyper Text Markup Language

What is XHTML?

- XHTML is a stricter, more XML-based version of HTML.
- XHTML stands for E**X**tensible **H**yper **T**ext **M**arkup **L**anguage.
- XHTML was developed to make HTML more extensible and flexible to work with other data formats (such as XML).
- In addition, browsers ignore errors in HTML pages, and try to display the website even if it has some errors in the markup. So XHTML comes with a much stricter error handling.

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The Most Important Differences from HTML

- <!DOCTYPE> is **mandatory**
- The xmlns attribute in <html> is **mandatory**
- <html>, <head>, <title>, and <body> are **mandatory**
- Elements must always be **properly nested**
- Elements must always be **closed**
- Elements must always be in **lowercase**
- Attribute names must always be in **lowercase**
- Attribute values must always be **quoted**
- Attribute minimization is **forbidden**

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XHTML - <!DOCTYPE ....> Is Mandatory

- An XHTML document must have an XHTML <!DOCTYPE> declaration.
- The <html>, <head>, <title>, and <body> elements must also be present, and the xmlns attribute in <html> must specify the xml namespace for the document.

Sample XHTML

```
<!DOCTYPE html >
<html xmlns=" http://www.w3.org/1999/xhtml ">
<head>
<title>Title of document</title>
</head>
<body>
   <p> Subject name: Foundations of Computing </p>
</body>
</html>
```

XHTML Elements Must be Properly Nested

- In XHTML, elements must always be properly nested within each other, like this,

Correct:

  Some text

Wrong:

Some text

XHTML Elements Must Always be Closed

- In XHTML, elements must always be closed, like this:

Correct:

```
<p>This is a paragraph</p>

<p>This is another paragraph</p>
```

Wrong:

```
<p>This is a paragraph

<p>This is another paragraph
```

XHTML Empty Elements Must Always be Closed

- In XHTML, empty elements must always be closed, like this:

Correct:

A break:

A horizontal rule: <hr />

An image: <img src=" rose.jpg alt=" ROSE FLOWER " />

Wrong:

A break:

A horizontal rule: <hr>

An image: <img src="rose.jpg" alt="ROSE FLOWER">

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Example

```
<!DOCTYPE html>
<html xmlns="http://www.w3.org/1999/xhtml">
<head>
   <title>XHTML Line Break and Horizontal Rule Example</title>
</head>
<body>

   <h1>Welcome to XHTML</h1>

   <p>This is a paragraph with a line break.<br />Here is the next line.</p>

   <hr />

   <p>This is another paragraph after the horizontal rule.</p>

</body>
</html>
```

XHTML Elements Must be in Lowercase

- In XHTML, element names must always be in lowercase, like this:

Correct:

```
<body>
<p>This is a paragraph</p>
</body>
```

Wrong:

```
<BODY>
<P>This is a paragraph</P>
</BODY>
```

XHTML Attribute Names Must be in Lowercase

- In XHTML, attribute names must always be in lowercase, like this:

Correct:

```
<a href="https://www.ktu.edu.in/">Visit our website</a>
```

Wrong:

```
<a HREF ="https://www.ktu.edu.in/"> Visit our website </a>
```

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XHTML Attribute Values Must be Quoted

Correct:

```
<a href="https://www.ktu.edu.in/">Visit our website</a>
```

Wrong

```
<a href=https://www.ktu.edu.in/>Visit our website</a>
```

Validating Your Web Content

- One very important way to test your pages is to *validate* them.
- Validation is the process of testing your pages with a special application that searches for errors and makes sure your pages follow the strict XHTML standard.
- To validate a page, follow this URL: http://validator.w3.org/. The W3C Markup Validation Service is shown in Figure 2.4.

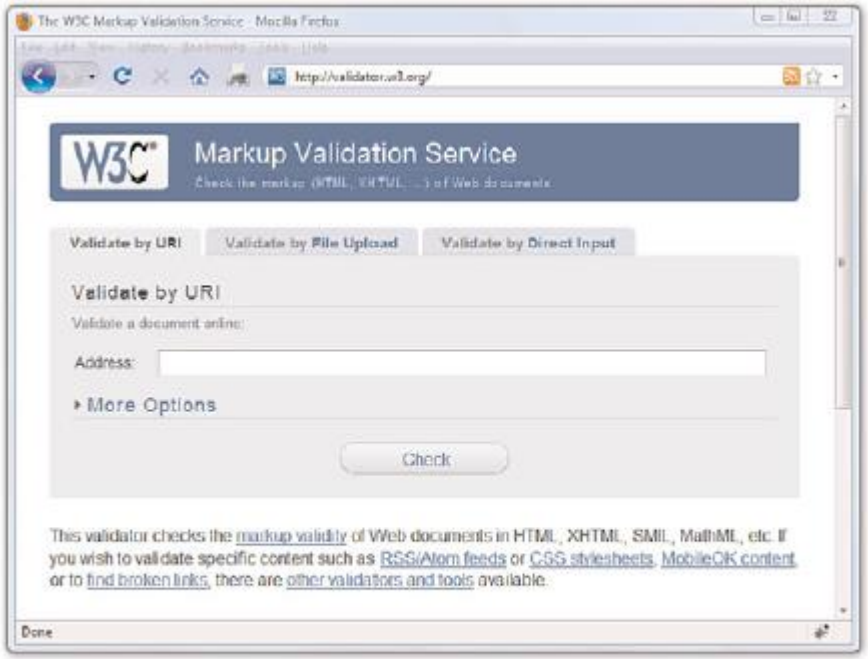

FIGURE 2.4
The W3C Markup Validation Service enables you to validate an HTML (XHTML) document to ensure it has been coded accurately.

The Scoop on HTML, XML, XHTML and HTML5

- HTML is a standard language used to create and design web pages. Main purpose of HTML is Structures content on the web — like headings, paragraphs, links, images, etc.

- XML is a general language used to create specific languages, such as HTML. It might sound a little strange, but it really just means that XML provides a basic structure and set of rules to which any markup language must adhere.

- Using XML, you can create a unique markup language to describe just about any kind of information, including web pages.

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- Using XML, you can create a unique markup language to describe just about any kind of information, including web pages.

The Scoop on HTML, XML, XHTML and HTML5

- HTML is a relatively unstructured language that benefits from the rules of XML. The natural merger of the two technologies resulted in HTML's adherence to the rules and structure of XML.

- The new XML-compliant version of HTML is known as XHTML. Combines the *structure of HTML* with the *strict rules of XML*.

- HTML 5, which is touted as the next web standard.
- HTML 5 is not a replacement for XHTML, but is a major revision of HTML 4. In other words, XHTML and HTML 5 can coexist on the web, and web browsers that currently support XHTML will also (one day) support HTML 5 as well.

- To modernize the web — adding support for video, audio, animations, better forms, offline browsing, and more without needing extra plugins like Flash.

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HTML vs XHTML

Parameter	HTML	XHTML
Full form	HTML stands for Hyper Text Markup Language	XHTML stands for extensible hypertext markup language
Developed by	HTML was developed by Tim Berners Lee	XHTML was developed by World Wide Web Consortium (W3C)
Release date	It was released in 1993	It was released in 2000
File format	It is of document type file format	It is in markup language file format
Extended from	HTML is extended from SGML (Standard Generalized Markup Language	It is extended from HTML and XML (Extensible markup language)
File extension	It has the file extensions - .html, .htm	It has the extensions - .xhtml, .xht, .html, .htm, .xml
Case sensitivity	HTML is not case sensitive. It can be written in upper case or lowercase or a combination of both	XHTML is case sensitive. All the code should be in lower case
DOCTYPE	In HTML, it is not necessary to add < DOCTYPE > at the beginning of every page	In XHTML, it is compulsory to mention < DOCTYPE > at the top of every page
Closing tags	In HTML, it is not mandatory to close the tags	It is mandatory to close all the tags in XHTML

Understanding Cascading Style Sheets

Cascading Style Sheets

- HTML was NEVER intended to contain tags for formatting a web page!
- HTML was created to describe the content of a web page
- When tags like <font>, and color attributes were added to the HTML 3.2 specification, it started a nightmare for web developers.
- Development of large websites, where fonts and color information were added to every single page, became a long and expensive process.
- To solve this problem, the World Wide Web Consortium (W3C) created CSS.
- CSS removed the style formatting from the HTML page!

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Cascading Style Sheets

- The technology behind style sheets is called CSS, which stands for Cascading Style Sheets.
- CSS is a language that defines style constructs such as fonts, colors, and positioning, which are used to describe how information on a web page is formatted and displayed.
- CSS is the language we use to style an HTML document.
- CSS describes how HTML elements should be displayed.

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Cascading Style Sheets

- You create a style sheet document that specifies the fonts, colors, spacing, and other characteristics that establish a unique look for a website.
- You then link every page that should have that look to the style sheet, instead of specifying all those styles repeatedly in each separate document.

- CSS styles can be stored directly in an HTML web page or in a separate style sheet file. Either way, style sheets contain style rules that apply styles to elements of a given type.
- When used externally, style sheet rules are placed in an external style sheet document with the file extension **.css**
- A *style rule* is a formatting instruction that can be applied to an element on a web page, such as a paragraph of text or a link. Style rules consist of one or more *style properties* and their associated values.

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CSS Demo - One HTML Page - Multiple Styles!

- Here we will show one HTML page displayed with four different stylesheets.

1. No Stylesheet

Welcome to My Homepage

Use the menu to select different Stylesheets

- Stylesheet 1
- Stylesheet 2
- Stylesheet 3
- Stylesheet 4
- No Stylesheet

Same Page Different Stylesheets

This is a demonstration of how different stylesheets can change the layout of your HTML page. You can change the layout of this page by selecting different stylesheets in the menu, or by selecting one of the following links:
Stylesheet1, Stylesheet2, Stylesheet3, Stylesheet4.

No Styles

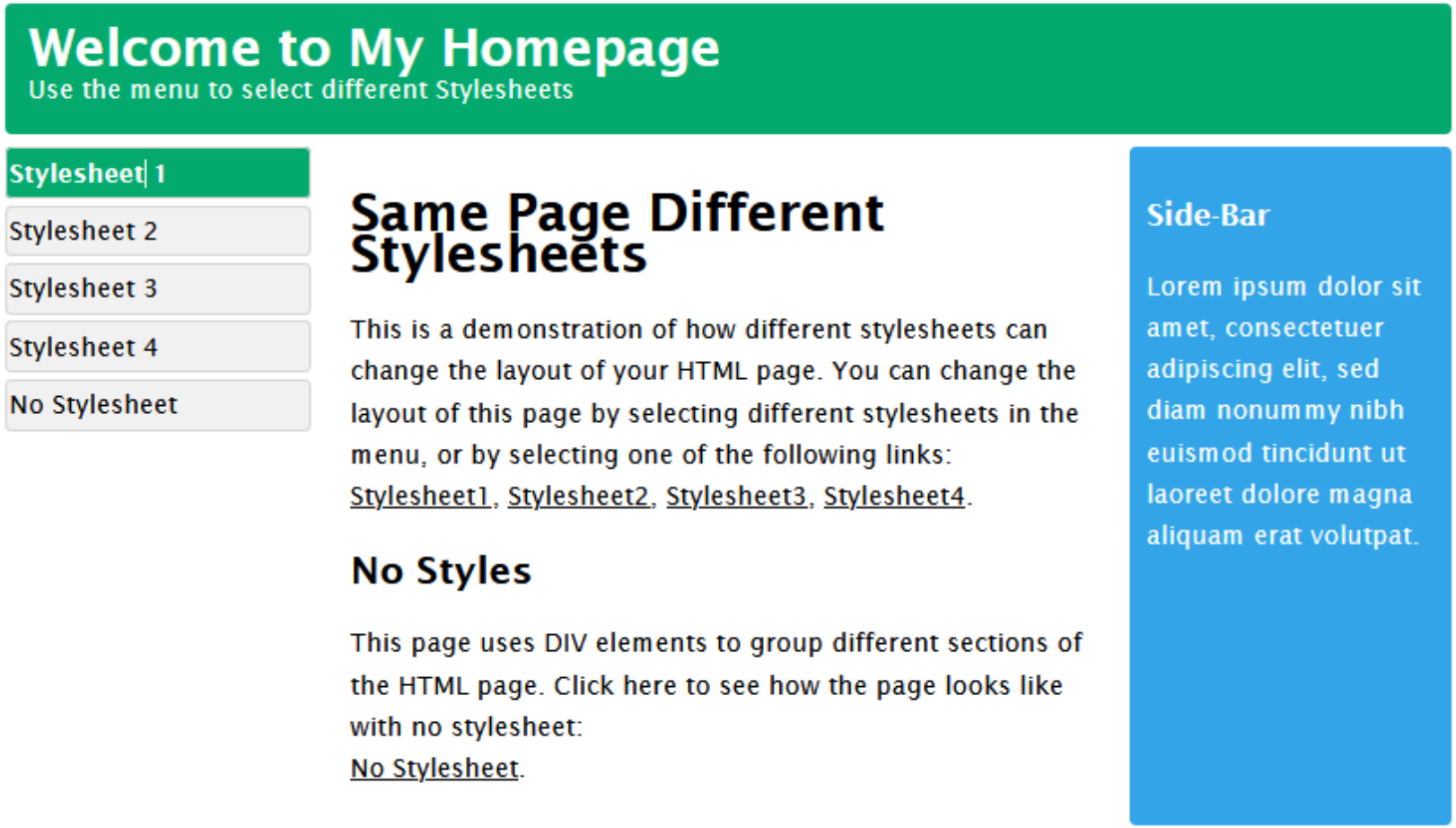
This page uses DIV elements to group different sections of the HTML page. Click here to see how the page looks like with no stylesheet:
No Stylesheet.

Side-Bar

Lorem ipsum dolor sit amet, consectetuer adipiscing elit, sed diam nonummy nibh euismod tincidunt ut laoreet dolore magna aliquam erat volutpat.

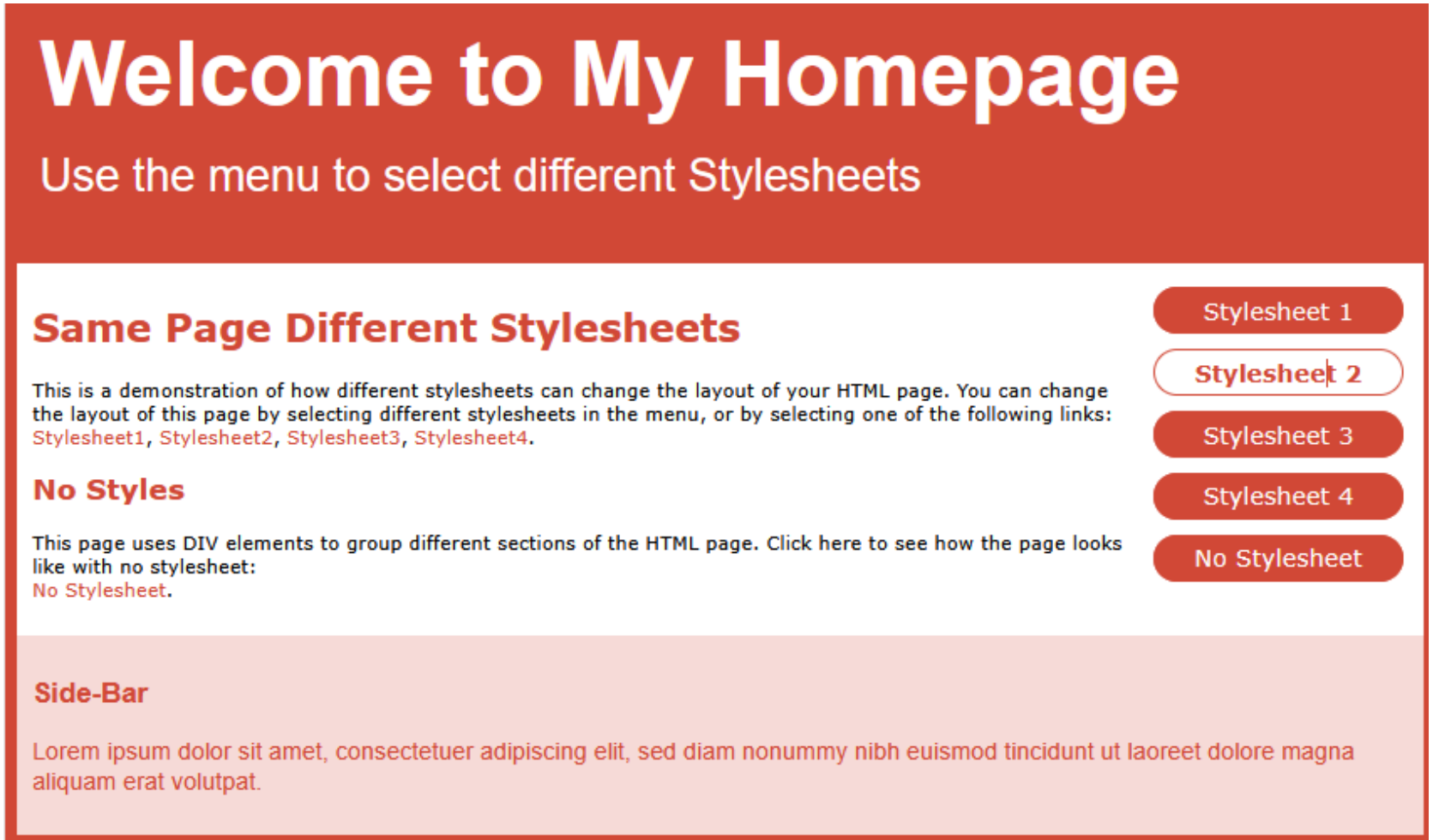
CSS Demo - One HTML Page - Multiple Styles!

- Stylesheet 1



CSS Demo - One HTML Page - Multiple Styles!

Stylesheet 2



How To Add CSS

Three Ways to Insert CSS

There are three ways of inserting a style sheet:

- External CSS
- Internal CSS
- Inline CSS

Inline CSS

- An inline style may be used to apply a unique style for a single element.
- To use inline styles, add the style attribute to the relevant element. The style attribute can contain any CSS property.

Example:

```
<!DOCTYPE html>
<html>
<body>

<h1 style="color:blue;text-align:center;">This is a heading</h1>
<p style="color:red;">This is a paragraph.</p>

</body>
</html>
```

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Output

This is a heading

This is a paragraph.

Internal CSS

- An internal style sheet may be used if one single HTML page has a unique style.
- The internal style is defined inside the <style> element, inside the head section.

Example

- Internal styles are defined within the <style> element, inside the <head> section of an HTML page:

```
<!DOCTYPE html>
<html>
<head>
<style>
body {
  background-color: linen;
}
```

Internal CSS

```
h1 {
  color: maroon;
  margin-left: 40px;
}
</style>
</head>

<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>

</html>
```

Output

External CSS

- With an external style sheet, you can change the look of an entire website by changing just one file!
- Each HTML page must include a reference to the external style sheet file inside the <link> element, inside the head section.

Example

- External styles are defined within the <link> element, inside the <head> section of an HTML page:

```
<!DOCTYPE html>
<html>
<head>
<link rel="stylesheet" href="mystyle.css">
</head>
<body>
```

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External CSS

```
<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

- An external style sheet can be written in any text editor, and must be saved with a .css extension.
- The external .css file should not contain any HTML tags.
- Here is how the "mystyle.css" file looks:

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External CSS(mystyle.css)

```
body {
  background-color: lightblue;
}

h1 {
  color: navy;
  margin-left: 20px;
}
```

Note: Do not add a space between the property value (20) and the unit (px):

Incorrect (space): margin-left: 20 px;

Correct (no space): margin-left: 20px;

Output

CSS Comments

- Comments are used to explain the code, and may help when you edit the source code at a later date.

- Comments are ignored by browsers.

- A CSS comment is placed inside the <style> element, and starts with /* and ends with */:

```
/* This is a single-line comment */
p {
  color: red;
}
```

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A CSS Style Primer

CSS Style Primer

- CSS includes various style properties that are used to control fonts, colors, alignment, and margins, to name just a few. The style properties in CSS can be generally grouped into two major categories:

- **Layout properties**—Consist of properties that affect the positioning of elements on a web page, such as margins, padding, alignment, and so on
- **Formatting properties**—Consist of properties that affect the visual display of elements within a website, such as the font type, size, color, and so on

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Layout Properties

- CSS layout properties are used to determine how content is placed on a web page. One of the most important layout properties is the display property, which describes how an element is displayed with respect to other elements.

There are four possible values for the display property:

- block—The element is displayed on a new line, as in a new paragraph.
- list-item—The element is displayed on a new line with a list-item mark (bullet) next to it.
- none—The element is not displayed; it is hidden.
- inline—The element is displayed inline with the current paragraph.

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Syntax

- `display: block;`
- `display: inline;`
- `display: none;`
- `display: list-item;`

Block level and inline elements

Here are the block-level elements in HTML:

<address>	<article>	<aside>	<blockquote>	<canvas>	<dd>	
<dl>	<dt>	<fieldset>	<figcaption>	<figure>	<footer>	<form>
<h1>-<h6>	<header>	<hr>	<li>	<main>	<nav>	<noscript>
<ol>	<p>	<pre>	<section>	<table>	<tfoot>	<ul>
<video>						

Here are the inline elements in HTML:

<a>	<abbr>	<acronym>	<b>	<bdo>	<big>	
<button>	<cite>	<code>	<dfn>		<i>	<img>
<input>	<kbd>	<label>	<map>	<object>	<output>	<q>
<samp>	<script>	<select>	<small>			<sub>
<sup>	<textarea>	<time>	<tt>	<var>		

Layout Properties

Many size-related CSS properties, width and height property values can be specified in several different units of measurement:

- in—Inches
- cm—Centimeters
- mm—Millimeters
- px—Pixels
- pt—Points

CSS Padding

- The CSS padding properties are used to generate space around an element's content, inside of any defined borders.

- With CSS, you have full control over the padding. There are properties for setting the padding for each side of an element (top, right, bottom, and left).

- Padding - Individual Sides

- CSS has properties for specifying the padding for each side of an element:

- padding-top
- padding-right
- padding-bottom
- padding-left

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Example

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  border: 1px solid black;
  background-color: lightblue;
  padding-top: 50px;
  padding-right: 30px;
  padding-bottom: 50px;
  padding-left: 80px;
}
</style>
</head>
```

Example….

```
<body>

<h2>Using individual padding properties</h2>

<div>This div element has a top padding of 50px, a right padding of 30px, a bottom padding of 50px,
and a left padding of 80px.</div>

</body>
</html>
```

Output

Using individual padding properties

This div element has a top padding of 50px, a right padding of 30px, a bottom padding of 50px, and a left padding of 80px.

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CSS Margins

- The CSS margin properties are used to create space around elements, outside of any defined borders.

- With CSS, you have full control over the margins. There are properties for setting the margin for each side of an element (top, right, bottom, and left).
- Margin - Individual Sides

- CSS has properties for specifying the margin for each side of an element:

- margin-top
- margin-right
- margin-bottom
- margin-left

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Example

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  border: 1px solid black;
  margin-top: 100px;
  margin-bottom: 100px;
  margin-right: 150px;
  margin-left: 80px;
  background-color: lightblue;
}
</style>
</head>
```

Example…

```
<body>

<h2>Using individual margin properties</h2>

<div>This div element has a top margin of 100px, a right margin of 150px, a bottom margin of
100px, and a left margin of 80px.</div>

</body>
</html>
```

Output

Using individual margin properties

This div element has a top margin of 100px, a right margin of 150px, a bottom margin of 100px, and a left margin of 80px.

Formatting Properties

- CSS formatting properties are used to control the appearance of content on a web page, as opposed to controlling the physical positioning of the content.
- One of the most popular formatting properties is the border property, which is used to establish a visible boundary around an element with a box or partial box.

The following **border properties** provide a means of describing the borders of an element:

- border-width—The width of the border edge
- border-color—The color of the border edge
- border-style—The style of the border edge

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Formatting Properties

- border-left—The left side of the border
- border-right—The right side of the border
- border-top—The top of the border
- border-bottom—The bottom of the border
- border—All the border sides

- The border-width property is used to establish the width of the border edge. It is often expressed in pixels, as the following code demonstrates:

```
border-width:5px;
border-color:blue;
border-style:dotted;
```

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Formatting Properties

The border-style property can be set to any of the following values:

- solid—A single-line border
- double—A double-line border
- dashed—A dashed border
- dotted—A dotted border
- groove—A border with a groove appearance
- ridge—A border with a ridge appearance
- inset—A border with an inset appearance
- outset—A border with an outset appearance
- none—No border

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Example

```
<!DOCTYPE html>
<html>
<head>
<style>
h1 {
  border: 5px solid red;
}

h2 {
  border: 4px dotted blue;
}
```

Example

```
div {
  border: double;
}
</style>
</head>

<body>
<h1>A heading with a solid red border</h1>

<h2>A heading with a dotted blue border</h2>
```

Example

```
<div>A div element with a double border.</div>

</body>
</html>
```

Output

A heading with a solid red border

A heading with a dotted blue border

A div element with a double border.

Color property

- The color of an element's border is set with the border-color property, the color of the inner region of an element is set using the color and background-color properties.
- The color property sets the color of text in an element (foreground) and the background-color property sets the color of the background behind the text.

Syntax

color:black;

background-color:orange;

Text align property

- You can also control the alignment and indentation of web page content without too much trouble. This is accomplished with the text-align and text-indent properties, as the following code demonstrates:
- **Syntax**

```
text-align:center;

text-indent:12px;
```

Font property

The following font properties are used to set the various parameters associated with fonts:

- font-family—The family of the font
- font-size—The size of the font
- font-style—The style of the font (normal or italic)
- font-weight—The weight of the font (light, medium, bold, and so on)

Syntax:

- font-family: Arial, sans-serif;
- font-size: 36pt;
- font-style: italic;
- font-weight: medium;

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Font property

- **text-decoration**—Useful for turning link underlining off—simply set it to none. The values of underline, italic, and line-through are also supported.

- **padding**—Adds padding to the left, right, top, and bottom of an element; this padding can be in measurement units or a percentage of the page width.
- Use padding-left and padding-right if you want to add padding to the left and right of the element independently.

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Practice questions- 1

Create a web page with title ,heading, image and add basic styles.

Answer

```
<!DOCTYPE html>
<html >
<head>
 <title>My Simple Webpage</title>
 <style>
        /* Basic CSS Styling */
        body {
            background-color: red;
            font-family: Arial, sans-serif;
            text-align: center;
            padding: 20px;
        }
```

```
h1 {
        color: blue;
        margin-bottom: 20px;
    }

img {
        width: 300px;
        height: auto;
        border: 5px solid green;
        margin-top: 20px;
    }
  </style>
```

```
</head>
<body>

    <h1>Welcome to My Webpage!</h1>

    <img src=“cat.jpg" alt="Sample Image">

</body>
</html>
```

Practice questions- 2

- Design a web page using HTML and CSS for an online dress shopping. Include title, heading, a paragraph which describe about the webpage and list of items, and basic style.

Answer

```
<!DOCTYPE html>
<html>
<head>

    <title>Dream Dresses - Online Store</title>

    <style>
body {
            font-family: 'Arial';
            background-color: red;
            color: black;
            /*padding: 20px;*/
            text-align: center;
        }
```

Answer

```
h1 {
        color: black;
        font-size: 40px;
        margin-bottom: 10px;
    }

    p {
        font-size: 18px;
        margin-bottom: 30px;
    }
```

Answer

```
li {
        background-color: #ffe4e1;
        margin: 10px auto;
        padding: 10px;
        width: 300px;
        /*border-radius: 8px;*/
        font-size: 20px;
    }
  </style>
</head>
```

Answer

```
<body>
    <h1>Dream Dresses</h1>

    <p>Welcome to Dream Dresses — your ultimate destination for stylish, trendy, and affordable dresses for every occasion. Browse our latest collection and find your perfect outfit today!</p>

    <h2>Our Collection:</h2>
    <ul>
        <li>Summer Floral Dress</li>
        <li>Elegant Evening Gown</li>
        <li>Casual Shirt Dress</li>
        <li>Party Wear Mini Dress</li>
        <li>Bohemian Maxi Dress</li>
    </ul>
```

Answer

```
</body>
</html>
```

Output

Dream Dresses

Welcome to Dream Dresses — your ultimate destination for stylish, trendy, and affordable dresses for every occasion. Browse our latest collection and find your perfect outfit today!

Our Collection:

- Summer Floral Dress
- Elegant Evening Gown
- Casual Shirt Dress
- Party Wear Mini Dress
- Bohemian Maxi Dress

CSS Selectors

CSS Selectors

- CSS selectors are used to "find" (or select) the HTML elements you want to style.

Using Style Classes (The CSS class Selector)

- Whenever you want some of the text on your pages to look different from the other text, you can create what amounts to a custom built HTML tag. Each type of specially formatted text you define is called a style class.
- A style class is a custom set of formatting specifications that can be applied to any element in a web page.
- The class selector selects HTML elements with a specific class attribute.
- To select elements with a specific class, write a period (.) character, followed by the class name.

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Example 1

```
<!DOCTYPE html>
<html>
<head>
<style>
.center {
  text-align: center;
  color: red;
}
</style>
</head>
```

Example 1

```
<body>

<h1 class="center">Red and center-aligned heading</h1>
<p class="center">Red and center-aligned paragraph.</p>

</body>
</html>
```

Output

Red and center-aligned heading

Red and center-aligned paragraph.

Example 2

```
<!DOCTYPE html>
<html>
<head>
<style>
p.center {
  text-align: center;
  color: red;
}
</style>
</head>
```

Example 2

```
<body>

<h1 class="center">This heading will not be affected</h1>
<p class="center">This paragraph will be red and center-aligned.</p>

</body>
</html>
```

Output

This heading will not be affected

This paragraph will be red and center-aligned.

Using Style IDs

- A style ID is a custom set of formatting specifications that can be applied only to one element in a web page. You can use IDs across a set of pages but only once per time within each page.

- The id selector uses the id attribute of an HTML element to select a specific element.

- The id of an element is unique within a page, so the id selector is used to select one unique element!

- To select an element with a specific id, write a hash (#) character, followed by the id of the element.

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Example

```
<!DOCTYPE html>
<html>
<head>
<style>
#para1 {
  text-align: center;
  color: red;
}
</style>
</head>
```

Example

```
<body>

<p id="para1">Hello World!</p>
<p>This paragraph is not affected by the style.</p>

</body>
</html>
```

Output

Hello World!

This paragraph is not affected by the style.

Summary

Tag/Attributes	Function
<style>...</style>	Allows an internal style sheet to be included within a document. Used between <head> and </head>.
Attribute	
type="contenttype"	The Internet content type. (Always "text/css" for a CSS style sheet.)
<link />	Links to an external style sheet (or other document type). Used in the <head> section of the document.
Attribute	
href="url"	The address of the style sheet.
type="contenttype"	The Internet content type. (Always "text/css" for a CSS style sheet.)
rel="stylesheet"	The link type. (Always "stylesheet" for style sheets.)
...	Does nothing but provide a place to put style or other attributes. (Similar to ... but does not cause a line break.)
Attribute	
style="style"	Includes inline style specifications. (Can be used in , , <body>, and most other HTML tags.)

Practice questions….(Class selectors)

1. Make a "card" layout.

- Create a with class card.
- Inside, add a heading and a paragraph.
- Style the card with a border, padding, and background color.1.

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Code….

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Card Layout</title>

    <style>
```

Code….

```
<style>
    /* Style for the card */
    div.card {
        width: 300px;
        margin: 50px auto;
        padding: 20px;
        background-color: #5117c5;
        border: 2px solid #ddd;
        border-radius: 10px;
        box-shadow: 0 2px 8px rgba(0, 0, 0, 0.1);
        font-family: Arial, sans-serif;
        text-align: center;
    }
```

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Code….

```
 /* Style for the card heading */
        h2.card  {
            margin-top: 0;
            color: #333;
        }
/* Style for the card paragraph */
        p.card {
            color: #666666;
            font-size: 16px;
        }
    </style>

</head>
<body>
```

Code….

```
<div class="card">
        <h2 class="card">Welcome to Our Site!</h2>
        <p calss="card">We offer the best services to help your business grow and succeed. 🚀</p>
    </div>


</body>
</html>
```

Practice questions ….(Class selectors)

2. Create 3 heading tags (<h1>) with different styles using class selectors.

One heading should be bold and blue.

One heading should be italic and green.

One heading should be underlined and red.

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Code…

```
<!DOCTYPE html>
<html>
<head>
<title>Styled Headings</title><style>

        .blue-bold {
            color: blue;
            font-weight: bold;
        }
```

Code…

```
.green-italic {
        color: green;
        font-style: italic;
    }

    .red-underline {
        color: red;
        text-decoration: underline;
    }
  </style>
</head>
```

Code…

```
<body>

    <h1 class="blue-bold">This is a Bold and Blue Heading</h1>
    <h1 class="green-italic">This is an Italic and Green Heading</h1>
    <h1 class="red-underline">This is an Underlined and Red Heading</h1>

</body>
</html>
```

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Practice questions ….(Class selectors)

3. Design a "Service Section".

- Create three s with classes like .service1, .service2, .service3.
- Each should have a title and description.
- Use different background colors and some margin between them.

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```
<!DOCTYPE html>
<html>
<head>
  <title>Service Section</title>
<style>
.service {
        padding: 20px;
        margin: 15px;
        border-radius: 10px;
        color: white;
        font-family: Arial, sans-serif;
      }
```

```
.service1 {
    background-color: #4CAF50; /* Green */
}

.service2 {
    background-color: #2196F3; /* Blue */
}

.service3 {
    background-color: #FF5722; /* Orange */
}
```

```
h2 {
        margin-top: 0;
      }
   </style>

</head>
<body><h1>Our Services</h1>

   <div class="service1">
      <h2>Web Development</h2>
      <p>We create responsive and dynamic websites tailored to your needs.</p>
   </div>
```

```
<div class="service2">
        <h2>Graphic Design</h2>
        <p>Our team crafts stunning visuals that communicate your brand story.</p>
    </div>

    <div class="service3">
        <h2>Digital Marketing</h2>
        <p>Grow your business online with our expert marketing strategies.</p>
    </div>

</body>
</html>
```

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Understanding JavaScript

Learning Web Scripting Basics

- HTML is a simple text markup language, it can't respond to the user, make decisions, or automate repetitive tasks.
- Interactive tasks such as these require a more sophisticated language: a programming language, or a scripting language.
- Scripting languages are generally simple. They have a simple syntax, can perform tasks with a minimum of commands, and are easy to learn.
- JavaScript, on the other hand, is an interpreted language: The browser executes each line of script as it comes to it.

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Introducing JavaScript

- JavaScript was developed by Netscape Communications Corporation, the maker of the Netscape web browser.
- JavaScript was the first web scripting language to be supported by browsers, and it is still by far the most popular.
- JavaScript is almost as easy to learn as HTML, and it can be included directly in HTML documents.
- Here are a few of the things you can do with JavaScript:

➢ Display messages to the user as part of a web page, in the browser's status line, or in alert boxes.

➢ Validate the contents of a form and make calculations.

➢ Animate images or create images that change when you move the mouse over them

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Introducing JavaScript

- Create ad banners that interact with the user, rather than simply displaying a graphic.
- Modify all or part of a web page without requiring the user to reload it
- Detect the browser in use or its features and perform advanced functions only on browsers that support them

How JavaScript Fits into a Web Page

- Using the **<script>** tag, you can add a short script (in this case, just one line) to a web document.
- The <script> tag tells the browser to start treating the text as a script, and the closing **</script>** tag tells the browser to return to HTML mode.

There are actually four different places where you might use scripts:

1. **Inside the <head> section** : The script runs before the page content is loaded.

```
<head>
  <script src="script.js"></script>
</head>
```

How JavaScript Fits into a Web Page

2. Inside the <body> section

- The script runs while the page is loading. The script's output can directly affect the page content.

```
<body>
  <h1>Welcome!</h1>
  <script>
    document.write("This is dynamic content!");
  </script>
</body>
```

How JavaScript Fits into a Web Page

3. **External file linked by <script src="...">**

- Instead of writing the script directly into the HTML, you link an external .js file. Keeps your HTML cleaner and separates structure (HTML) from behavior (JavaScript).

```
<script src="external-script.js"></script>
```

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How JavaScript Fits into a Web Page

4 Inline event handlers : Small pieces of JavaScript can be added directly inside HTML elements. For example, handling a button click:

```
<button onclick="alert('Hello!')">Click me</button>
```

<button> This is an HTML element that creates a clickable button on the page.

onclick="..." This is an event handler attribute. It tells the browser: "When the button is clicked, run the JavaScript code inside the quotes.“

alert('Hello!') This is the JavaScript code inside the onclick. It calls the alert() function, which shows a popup box with the message 'Hello!'.

Click me This is the text that appears on the button itself.

</button> This closes the button element.

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Using Separate JavaScript Files

- External scripts are supported by all modern browsers. To use an external script, you specify its filename in the <script> tag:

```
<script src=”filename.js”></script>
```

You can create the .js file using a text editor. It should contain one or more JavaScript commands and only JavaScript—don’t include <script> tags other HTML tags, or HTML comments. Save the .js file in the same director as the HTML documents that refer to it.

Understanding JavaScript Events

- You can do with JavaScript involve interacting with the user and that means responding to events—for example, a link or a button being clicked. You can define event handlers within HTML tags to tell the browser how to respond to an event.

Simple event handler

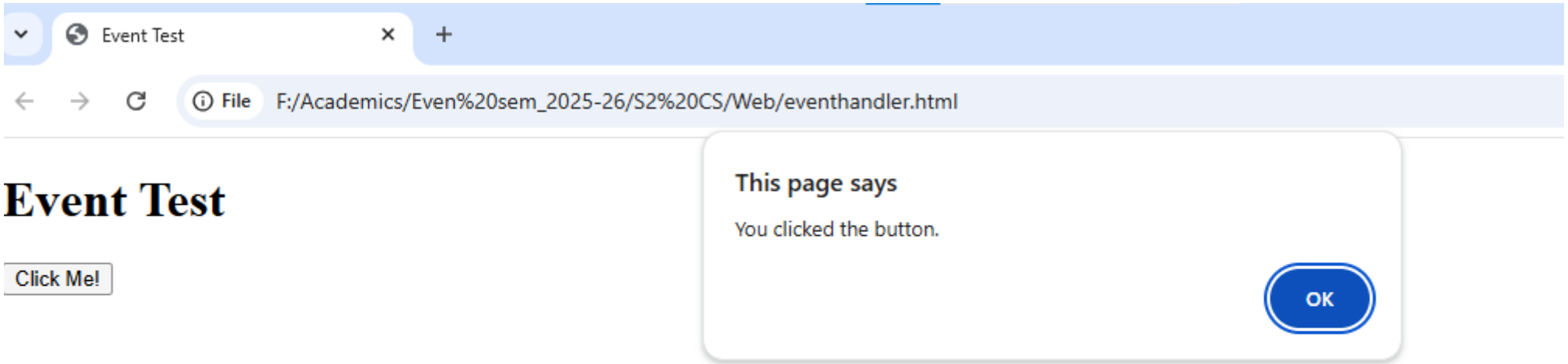
```
<!DOCTYPE html >
<html>
<head>
<title>Event Test</title>
</head>
<body>
<h1>Event Test</h1>

<button onclick="alert('You clicked the button.')">

Click Me!</button>
</body>
</html>
```

Simple event handler

- **onclick** is an HTML event attribute. It tells the browser: "When the user clicks on this element, run some JavaScript.“

- alert('You clicked the button.') is the JavaScript that gets run. It pops up a little alert box with the message "You clicked the button."

Output


Event Test
File
F:/Academics/Even%20sem_2025-26/S2%20CS/Web/eventhandler.html
Event Test
Click Me!
This page says
You clicked the button.
OK

Exploring JavaScript's Capabilities

Exploring JavaScript's Capabilities(Application)

- **Improving Navigation :** Some of the most common uses of JavaScript are in navigation systems for websites.
- You can use JavaScript to create a navigation tool—for example, a drop-down menu to select the next page to read or a submenu that pops up when you hover over a navigation link.

- **Validating Forms :** Form validation is another common use of JavaScript. A simple script can read values the user types into a form and can make sure they're in the right format.
- A simple script can read values the user types into a form and can make sure they're in the right format, such as email id, or phone numbers.

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Exploring JavaScript's Capabilities(Application)

- **Special Effects :** One of the earliest and most annoying uses of JavaScript was to create attention-getting special effects—for example, scrolling a message in the browser's status line or flashing the background color of a page.

- **Remote Scripting (AJAX) :** For a long time, the biggest limitation of JavaScript was that there was no way for it to communicate with a web server.
- For example, you could use it to verify that a phone number had the right number of digits, but not to look up the user's location in a database based on the number.
- Now that some of JavaScript's advanced features are supported by most browsers, this is no longer the case. Your scripts can get data from a server without loading a page or send data back to be saved. These features are collectively known as AJAX (Asynchronous JavaScript And XML), or remote scripting.

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Displaying Time with JavaScript

- One common and easy use for JavaScript is to display dates and times. Because JavaScript runs on the browser, the times it displays will be in the user's current time zone. However, you can also use JavaScript to calculate "universal" (UTC) time.

Beginning the Script

- Begins with the HTML <script> tag.
- Use the <script> and </script> tags to enclose a script within the HTML document.
- To begin creating the script, open your favorite text editor and type the beginning and ending <script> tags as shown.

```
<script> </script>
```

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Adding JavaScript Statements

Storing Data in Variables – date & time

- To begin the script, you will use a variable to store the current date.

- A variable is a container that can hold a value—a number, some text, or in this case, a date.

- To start writing the script, add the following line after the first <script> tag. Be sure to use the same combination of capital and lowercase letters in your version because JavaScript commands and variable names are case sensitive.

```
now = new Date();
```

This statement creates a **variable called now** and **current date and time** in it.

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Calculating the Results

- JavaScript includes a number of functions to convert dates and times in various ways.
- To continue your script, add the following two statements before the final </script> tag:

```
localtime = now.toString();
utctime = now.toGMTString();
```

These statements create two new variables: localtime, containing the current time and date in a nice readable format, and utctime, containing th UTC equivalent.

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GMT/UTC : (Greenwich Mean Time / Coordinated Universal Time)

Creating Output

- JavaScript includes a number of ways to display information, and one of the simplest is the **document.write** statement.
- The document.write statement displays a text string, a number, or anything else you throw at it.

```
document.write("<strong>Local time:</strong> " + localtime + "<br/>");

document.write("<strong>UTC time:</strong> " + utctime);
```

- The output will include some brief strings introducing the results and the contents of the localtime and utctime variables.

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Adding the Script to a Web Page

```
<!DOCTYPE html >
<head>
<title>Displaying Times and Dates</title>
</head>
<body>
<h1>Current Date and Time</h1>
<script>
now = new Date();
localtime = now.toString();
utctime = now.toGMTString();
```

Adding the Script to a Web Page

```
document.write("<strong>Local time:</strong> " + localtime + "<br/>");
document.write("<strong>UTC time:</strong> " + utctime);
</script>
</body>
</html>
```

Output

Current Date and Time

Local time: Wed Apr 09 2025 20:15:50 GMT+0530 (India Standard Time)
UTC time: Wed, 09 Apr 2025 14:45:50 GMT

Adding the Script to a Web Page

- To display a large clock, we need the hours, minutes, and seconds in separate variables. Once again, JavaScript has built-in functions to do most of the work:

```
hours = now.getHours();
mins = now.getMinutes();
secs = now.getSeconds();
```

These statements load the hours, mins, and secs variables with the component of the time using JavaScript's built-in date functions.

Adding the Script to a Web Page

- The first statement displays an HTML <h1> header tag to display the clock in a large typeface. The second statement displays the hours, mins, and secs variables, separated by colons, and the third adds the closing </h1> tag.

The Date and Time Script with Large Clock Display

```
<!DOCTYPE html >
<head>
<title>Displaying Times and Dates</title>
</head>
<body>
<h1>Current Date and Time</h1>
<script>
now = new Date();
localtime = now.toString();
utctime = now.toGMTString();
```

The Date and Time Script with Large Clock Display

```
document.write("<strong>Local time:</strong>"  + localtime + "<br/>");
document.write("<strong>UTC time:</strong> " + utctime);
hours = now.getHours();
mins = now.getMinutes();
secs = now.getSeconds();
document.write("<h1>");
document.write(hours + ":" + mins + ":" + secs);
document.write("</h1>");
</script>
</body>
</html>
```

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Output

Current Date and Time

Local time: Wed Apr 09 2025 20:32:22 GMT+0530 (India Standard Time)
UTC time: Wed, 09 Apr 2025 15:02:22 GMT

20:32:22

Practice question

1. Write a java script to display current date and time with basic styles.

2. Add a millisecond field to the large clock. You can use the getMilliseconds function, which works just like getSeconds but returns milliseconds.(Output sample)

11:52:52:402

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3. Modify the script to display the time, including milliseconds, twice. Notice whether any time passes between the two time displays when you load the page.

Answer 3 : Code…

```
<!DOCTYPE html>
<html >
<head>
  <title>Displaying Times and Dates Twice</title>
</head>
<body>
  <h1>Current Date and Time</h1>
```

```
<script>
    // First time capture
    now1 = new Date();
    localtime1 = now1.toString();
    utctime1 = now1.toGMTString();
    hours1 = now1.getHours();
    mins1 = now1.getMinutes();
    secs1 = now1.getSeconds();
    millis1 = now1.getMilliseconds();
```

```
document.write("<strong>First Local Time:</strong> " + localtime1 + "<br/>");
document.write("<strong>First UTC Time:</strong> " + utctime1 + "<br/>");
document.write("<h2>");
document.write(hours1 + ":" + mins1 + ":" + secs1 + ":" + millis1);
document.write("</h2><br/>");

// Second time capture

now2 = new Date();
localtime2 = now2.toString();
utctime2 = now2.toGMTString();
hours2 = now2.getHours();
mins2 = now2.getMinutes();
secs2 = now2.getSeconds();
millis2 = now2.getMilliseconds();
```

```
  document.write("<strong>Second Local Time:</strong> " + localtime2 + "<br/>");
  document.write("<strong>Second UTC Time:</strong> " + utctime2 + "<br/>");
  document.write("<h2>");
  document.write(hours2 + ":" + mins2 + ":" + secs2 + "." + millis2);
  document.write("</h2>");
 </script>

</body>
</html>
```

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Testing the Script

- To test your script, you simply need to load the HTML document you created in a web browser. If you typed the script correctly, your browser should display the result of the script.

- Depending on your security settings, the script might not execute, and a yellow highlighted bar on the top of the browser might display a security warning. In this case, click the yellow bar and select Allow Blocked Content to allow your script to run.

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Dealing with JavaScript Errors

- As you develop more complex JavaScript applications, you're going to run into errors from time to time. JavaScript errors are usually caused by mistyped JavaScript statements.

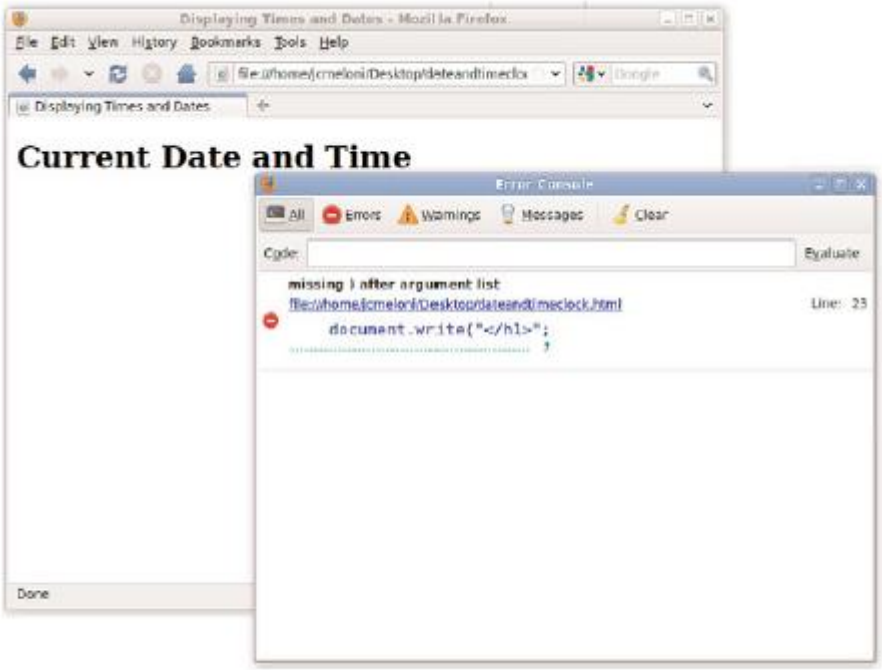
- We'll use a common error: omitting one of the parentheses. Change the last document.write statement in the previous code, to read:

  document.write("</h1>";

- Save your HTML document again and load the document into the browser.
- Depending on the browser version you're using, one of two things will happen: Either an error message will be displayed, or the script will simple fail to execute.

Dealing with JavaScript Errors

- If an error message is displayed, you're halfway to fixing the problem by adding the missing parenthesis. If no error was displayed, you should configure your browser to display error messages so that you can diagnose future problems:
- In Firefox, you can also select Tools, JavaScript Console from the menu. The console is shown in Figure 4.4, displaying the error message you created in this example.



Dealing with JavaScript Errors

- In Chrome, select Tools, JavaScript Console from the Customizations (Options) menu. A console will display in the bottom of the browser window.

- In Internet Explorer, select Tools, Internet Options. On the Advanced page, uncheck the Disable Script Debugging box and check the Display a Notification About Every Script Error box. (If this is disabled, a yellow icon in the status bar will still notify you of errors.)

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